

Team Biweekly Report Summary

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Period: May 26 - June 15, 2025

Progress Overview

The team has successfully completed all remaining project deliverables during this final reporting period. All tasks including user acceptance testing, system deployment, debugging, official launch, comprehensive documentation, and project closure activities have been finalized with 100% completion status. The system is now fully operational and has been successfully handed over to the operations team. This marks the successful completion of the Smart Maintenance Platform project.

Project Progress Status

According to the project Gantt chart, during this reporting period (May 26 - June 15, 2025), the team has completed the following WBS tasks:

- **Task 5.4 (User Acceptance Testing):** 100% complete (Fubin Chen)
- **Task 6.1 (Deployment Environment):** 100% complete (Jingxiao Han)
- **Task 6.2 (System Deploy):** 100% complete (All team members)
- **Task 6.3 (System Debugging):** 100% complete (All team members)
- **Task 6.4 (Official Launch):** 100% complete (Xuanhe Yang)
- **Task 7.1 (System Documentation Finalization):** 100% complete (All team members)
- **Task 7.2 (Project Documentation Archiving):** 100% complete (All team members)
- **Task 7.3 (Lessons Learned Documentation):** 100% complete (All team members)
- **Task 7.4 (Project Closure Report):** 100% complete (Xuanhe Yang)

Current project progress: 100% complete. The team has successfully completed all development, testing, deployment, documentation, and closure tasks. The project has been officially closed and delivered.

Team Contributions in Relation to WBS

Xuanhe Yang (30 hours):

- WBS 6.2: System Deploy (3 hours) - Deployment coordination and validation
- WBS 6.3: System Debugging (4 hours) - Database and data processing issues
- WBS 6.4: Official Launch (4 hours) - 100% Complete
- WBS 7.1: System Documentation Finalization (7.5 hours) - Testing and quality documentation
- WBS 7.2: Project Documentation Archiving (5 hours) - Project management documents
- WBS 7.3: Lessons Learned Documentation (5 hours) - Project management experiences
- WBS 7.4: Project Closure Report (1.5 hours) - Final project closure documentation

Fubin Chen (30 hours):

- WBS 5.4: User Acceptance Testing (12 hours) - 100% Complete
- WBS 6.2: System Deploy (3 hours) - Frontend deployment and validation
- WBS 6.3: System Debugging (4 hours) - Frontend and user experience issues
- WBS 7.1: System Documentation Finalization (7.5 hours) - User manuals and guides
- WBS 7.2: Project Documentation Archiving (3.5 hours) - User documentation archiving

Tong Li (27 hours):

- WBS 6.2: System Deploy (3 hours) - AI model deployment and configuration
- WBS 6.3: System Debugging (4 hours) - AI model integration and prediction functions
- WBS 7.1: System Documentation Finalization (7.5 hours) - System architecture and design documentation
- WBS 7.2: Project Documentation Archiving (2.5 hours) - Technical documentation archiving

- WBS 7.3: Lessons Learned Documentation (10 hours) - Testing and quality control experiences

Jingxiao Han (25 hours):

- WBS 6.1: Deployment Environment (2 hours) - 100% Complete
- WBS 6.2: System Deploy (3 hours) - Backend services deployment
- WBS 6.3: System Debugging (4 hours) - Backend services and API interfaces
- WBS 7.1: System Documentation Finalization (7.5 hours) - Technical and API documentation
- WBS 7.2: Project Documentation Archiving (3.5 hours) - Technical documentation archiving
- WBS 7.3: Lessons Learned Documentation (5 hours) - Technical implementation experiences

Key Deliverables

1. **Complete System Deployment and Launch:**
 - Production environment successfully configured and validated
 - System deployed with zero-downtime migration strategy
 - All system components verified and operational
 - Official system launch completed with stakeholder approval
2. **Final User Acceptance Testing:**
 - Final UAT report with stakeholder sign-off
 - All critical and high-priority issues resolved
 - User training materials and guides completed
 - System performance validation in production environment
3. **Comprehensive Documentation Package:**
 - Complete system architecture and design documentation
 - User manuals and operation guides
 - Technical documentation and API references
 - Testing documentation and quality assurance reports
 - All project documents properly archived
4. **Project Closure Documentation:**
 - Comprehensive lessons learned documentation
 - Project closure report with final metrics and outcomes
 - Knowledge transfer documentation for operations team
 - Project archive with all deliverables and artifacts

Final Project Metrics

- **Schedule Performance:** Project completed on time (June 15, 2025)
- **Budget Performance:** Within approved budget constraints
- **Quality Performance:** All deliverables met quality standards and stakeholder requirements
- **Scope Performance:** 100% of planned scope delivered successfully
- **Stakeholder Satisfaction:** High satisfaction ratings from all stakeholder groups

Challenges & Solutions

Challenge: Coordinating final deployment activities across multiple environments

- **Solution:** Implemented detailed deployment runbook with clear responsibilities and rollback procedures

Challenge: Ensuring comprehensive documentation while meeting tight deadlines

- **Solution:** Established parallel documentation streams and implemented peer review processes

Challenge: Managing final user acceptance criteria and sign-offs

- **Solution:** Created structured UAT checklists and facilitated dedicated stakeholder review sessions

Project Closure Summary

The Smart Maintenance Platform for Aero Engine Industrial Equipment project has been successfully completed on June 15, 2025. All planned deliverables have been delivered, tested, and deployed. The system is now operational and providing value to end users. Key success factors included:

1. Strong team collaboration and communication
2. Effective risk management and issue resolution
3. Comprehensive testing and quality assurance processes
4. Successful stakeholder engagement and change management
5. Thorough documentation and knowledge transfer

Resource Summary

- **Total Working Hours:** 112 hours (this period)
- **Total Project Hours:** 549 hours (100% utilized)
- **Schedule Status:** Completed on time
- **Project Health:** Excellent - project successfully closed with all objectives achieved

Individual Biweekly Reports

Biweekly Report

Name: Xuanhe Yang

Period: May 26 - June 15, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: System Debugging and Launch Preparation (May 26 - June 1, 2025)

During this week, I focused on system debugging activities and preparing for the official system launch:

1. **System Debugging (May 31 - June 1)**
 - Resolved database connection issues and data processing bottlenecks
 - Optimized AI model inference pipeline for production workloads
 - Fixed data synchronization issues between monitoring and prediction modules
 - Validated system performance under production-level data loads
2. **Official Launch Preparation (June 2 - June 3)**
 - Coordinated launch activities with all team members
 - Executed final system validation checks
 - Monitored system stability during initial production deployment
 - Documented launch procedures and post-launch monitoring protocols

Time spent: 8 hours

Week 2: Documentation and Knowledge Transfer (June 4 - June 8, 2025)

This week was dedicated to finalizing system documentation and capturing project lessons learned:

1. **System Documentation Finalization (June 4 - June 8)**
 - Completed testing documentation and quality assurance reports
 - Finalized system validation and performance benchmark documentation
 - Created troubleshooting guides for common system issues
 - Developed system monitoring and maintenance procedures
2. **Lessons Learned Documentation (June 4 - June 8)**
 - Documented project management experiences and team coordination insights
 - Captured stakeholder management and communication lessons

Time spent: 12.5 hours

Week 3: Project Closure Activities (June 9 - June 15, 2025)

This final week focused on completing all project closure activities:

1. **System Deploy Coordination (June 9 - June 10)**
 - Coordinated final deployment activities across all environments
 - Validated system stability and performance metrics
 - Ensured proper handover to operations team
2. **Project Documentation Archiving (June 11 - June 13)**
 - Organized and archived all project management documents
 - Created project repository with proper version control
 - Established access permissions for future reference
3. **Project Closure Report (June 14 - June 15)**
 - Compiled comprehensive project closure report
 - Documented final project metrics and outcomes
 - Prepared presentation for stakeholder closure meeting

Time spent: 9.5 hours

Completed WBS Items

- WBS 6.2: System Deploy (Coordination) - 100% Complete
- WBS 6.3: System Debugging - 100% Complete
- WBS 6.4: Official Launch - 100% Complete
- WBS 7.1: System Documentation Finalization (Testing & Quality) - 100% Complete
- WBS 7.2: Project Documentation Archiving (PM Documents) - 100% Complete
- WBS 7.3: Lessons Learned Documentation (Project Management) - 100% Complete
- WBS 7.4: Project Closure Report - 100% Complete

Challenges & Solutions

The primary challenge during this period was coordinating the complex system launch while ensuring all components remained stable. I addressed this by:

1. Creating a detailed launch checklist with clear go/no-go criteria
2. Establishing real-time monitoring dashboards for all critical system components

Final Accomplishments

Successfully led the project to completion with all deliverables met on time and within scope. Ensured smooth transition to operations team and comprehensive project closure documentation.

Total Hours Worked: 30 hours

Biweekly Report

Name: Jingxiao Han

Period: May 26 - June 15, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: Deployment Environment and System Debugging (May 27 - June 1, 2025)

During this week, I focused on deployment environment preparation and system debugging activities:

1. **Deployment Environment Setup (May 27 - May 28)**
 - Configured production server environment with required dependencies
 - Set up database connections and validated data migration procedures
 - Established system monitoring and logging infrastructure
 - Verified security configurations and access controls
2. **System Debugging (May 31 - June 1)**
 - Resolved backend service integration issues
 - Fixed API endpoint performance bottlenecks
 - Debugged user authentication and authorization problems
 - Optimized database connection pooling and query performance

Time spent: 6 hours

Week 2: Documentation and Knowledge Capture (June 4 - June 8, 2025)

This week was dedicated to technical documentation and lessons learned capture:

1. **Technical Documentation (June 4 - June 8)**
 - Completed API documentation with detailed endpoint specifications
 - Finalized technical architecture documentation
 - Created deployment and configuration guides
 - Developed troubleshooting procedures for common technical issues
2. **Technical Implementation Lessons (June 4 - June 8)**
 - Documented backend development best practices and patterns
 - Captured integration challenges and solutions

Time spent: 12.5 hours

Week 3: Final Deployment and Documentation Archiving (June 9 - June 15, 2025)

This final week focused on completing deployment activities and archiving technical documentation:

1. **Backend Services Deployment (June 9 - June 10)**
 - Executed final backend services deployment to production
 - Validated all API endpoints and service integrations
 - Monitored system performance and resolved any issues
2. **Technical Documentation Archiving (June 11 - June 15)**
 - Organized and archived all technical documentation
 - Created technical knowledge base for operations team
 - Established documentation maintenance procedures

Time spent: 6.5 hours

Completed WBS Items

- WBS 6.1: Deployment Environment - 100% Complete
- WBS 6.2: System Deploy (Backend Services) - 100% Complete
- WBS 6.3: System Debugging (Backend Services) - 100% Complete
- WBS 7.1: System Documentation Finalization (Technical & API) - 100% Complete

- WBS 7.2: Project Documentation Archiving (Technical Documentation) - 100% Complete
- WBS 7.3: Lessons Learned Documentation (Technical Implementation) - 100% Complete

Challenges & Solutions

The main challenge was ensuring seamless deployment in the production environment while maintaining system security. I addressed this by:

1. Implementing automated deployment scripts with validation checkpoints
2. Creating comprehensive security checklists for production deployment

Final Accomplishments

Successfully deployed all backend services and established a robust technical documentation archive for future maintenance and development activities.

Total Hours Worked: 25 hours

Biweekly Report

Name: Fubin Chen

Period: May 26 - June 15, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: User Acceptance Testing Completion (May 26 - June 1, 2025)

During this week, I focused on completing the user acceptance testing phase and supporting system debugging:

1. **User Acceptance Testing Completion (May 26 - May 30)**
 - Conducted final UAT sessions with all stakeholder groups
 - Validated all user workflows and interface functionality
 - Collected and prioritized final user feedback
 - Obtained formal sign-off from all stakeholder representatives
 - Created final UAT report with recommendations
2. **System Debugging Support (May 31 - June 1)**
 - Resolved frontend interface issues identified during final testing
 - Fixed user experience problems and interface inconsistencies
 - Optimized dashboard loading performance and responsiveness
 - Validated cross-browser compatibility across all supported platforms

Time spent: 16 hours

Week 2: User Documentation (June 4 - June 8, 2025)

This week was dedicated to finalizing user-facing documentation and training materials:

1. **User Documentation Finalization (June 4 - June 8)**
 - Completed comprehensive user manuals for all system roles
 - Created step-by-step operation guides with screenshots
 - Developed quick reference cards for common tasks
 - Finalized training materials and video tutorials
 - Created user onboarding checklists and procedures

Time spent: 7.5 hours

Week 3: Frontend Deployment and Documentation Archiving (June 9 - June 15, 2025)

This final week focused on frontend deployment and user documentation archiving:

1. **Frontend Deployment (June 9 - June 10)**
 - Deployed frontend components to production environment
 - Validated user interface functionality and performance
 - Conducted final user experience testing
2. **User Documentation Archiving (June 11 - June 15)**
 - Organized and archived all user documentation
 - Created user knowledge base and help system
 - Established procedures for documentation updates

Time spent: 6.5 hours

Completed WBS Items

- WBS 5.4: User Acceptance Testing - 100% Complete
- WBS 6.2: System Deploy (Frontend) - 100% Complete
- WBS 6.3: System Debugging (Frontend & UX) - 100% Complete
- WBS 7.1: System Documentation Finalization (User Manuals) - 100% Complete
- WBS 7.2: Project Documentation Archiving (User Documentation) - 100% Complete

Challenges & Solutions

The primary challenge was accommodating diverse user feedback while maintaining system consistency. I addressed this by:

1. Implementing configurable user interface options to meet different preferences
2. Creating role-based user guides tailored to specific user types
3. Developing a feedback prioritization matrix to focus on critical issues
4. Establishing clear criteria for determining which feedback to implement

Final Accomplishments

Successfully completed all user acceptance testing activities and established comprehensive user documentation that will support effective system adoption and usage.

Total Hours Worked: 30 hours

Biweekly Report

Name: Tong Li

Period: May 26 - June 15, 2025

Project: Smart Maintenance Platform for Aero Engine Industrial Equipment

Week 1: AI Model Debugging (May 31 - June 1, 2025)

During this week, I focused on debugging AI model integration and prediction functionality:

1. AI Model Integration Debugging (May 31 - June 1)

- Resolved model loading and initialization issues in production environment
- Fixed prediction pipeline performance bottlenecks
- Debugged data preprocessing and feature extraction problems
- Validated model output accuracy and consistency
- Optimized memory usage for large-scale model inference

Time spent: 4 hours

Week 2: Architecture Documentation and Quality Lessons (June 4 - June 8, 2025)

This week was dedicated to finalizing system architecture documentation and capturing quality control experiences:

1. System Architecture Documentation (June 4 - June 8)

- Completed detailed system architecture diagrams and specifications
- Finalized design decision documentation with rationale
- Created component interaction diagrams and data flow documentation
- Developed system maintenance and upgrade procedures
- Documented AI model architecture and integration patterns

2. Testing and Quality Control Lessons (June 4 - June 8)

- Documented comprehensive testing methodologies and best practices
- Captured quality assurance processes and validation procedures
- Created recommendations for future testing strategy improvements

Time spent: 17.5 hours

Week 3: AI Model Deployment and Documentation Archiving (June 9 - June 15, 2025)

This final week focused on AI model deployment and technical documentation archiving:

1. AI Model Deployment (June 9 - June 10)

- Deployed AI models to production environment
- Validated model performance and accuracy in production
- Established model monitoring and alerting systems

2. Technical Documentation Archiving (June 11 - June 15)

- Organized and archived all system architecture documentation
- Created technical design repository for future reference
- Established procedures for architecture documentation updates

Time spent: 5.5 hours

Completed WBS Items

- WBS 6.2: System Deploy (AI Model Deployment) - 100% Complete
- WBS 6.3: System Debugging (AI Model Integration) - 100% Complete
- WBS 7.1: System Documentation Finalization (Architecture & Design) - 100% Complete
- WBS 7.2: Project Documentation Archiving (Technical Documentation) - 100% Complete
- WBS 7.3: Lessons Learned Documentation (Testing & Quality) - 100% Complete

Challenges & Solutions

The main challenge was ensuring AI model reliability and performance in the production environment. I addressed this by:

1. Implementing comprehensive model validation and monitoring procedures
2. Creating automated testing frameworks for model accuracy verification

Final Accomplishments

Successfully deployed AI models to production and established comprehensive technical documentation that will support future system maintenance and enhancements.

Total Hours Worked: 27 hours